

Calculus II 第十三章

考点:空间曲线的参数方程, 求 $\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|}$, $\mathbf{N} = \frac{d\mathbf{T}/dt}{|d\mathbf{T}/dt|}$, $\kappa = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right| = \frac{|f''(x)|}{[1+(f'(x))^2]^{3/2}}$.

1. (2019年期中) Determine whether the following statements are true or false? No justification is necessary.

(4) If a vector function $\mathbf{r}(t)$ is always perpendicular to its derivative $\frac{d\mathbf{r}}{dt}$, then $|\mathbf{r}(t)|$ must be constant.

(5) The curvature of a unit circle is greater than the curvature of the parabola $y = x^2$ at the origin.

2. (2023年期末) Which of the following vector is orthogonal to the tangent line for the curve $\mathbf{r}(t) = (1 + \frac{1}{t^2})\mathbf{i} + (1 - \frac{3}{t})\mathbf{j} + t^2\mathbf{k}$ at the point $P = (2, -2, 1)$?

A. $\mathbf{v} = (1, 2, 1)$ B. $\mathbf{v} = (1, 2, 2)$ C. $\mathbf{v} = (-1, -2, 2)$ D. $\mathbf{v} = (-1, -2, -2)$

3. (2023年期中) The arc length of the curve $\mathbf{r}(t) = \frac{2}{3}t^{\frac{3}{2}}\mathbf{i} + \frac{2}{3}(2-t)^{\frac{3}{2}}\mathbf{j} + (t-1)\mathbf{k}$ for $\frac{1}{4} \leq t \leq \frac{1}{2}$ is ()

A. $\frac{1}{4}$ B. $\frac{\sqrt{2}}{4}$ C. $\frac{\sqrt{3}}{4}$ D. $\frac{1}{2}$

4. (2023年期中) A particle is traveling with acceleration $\mathbf{a}(t) = (e^t, t, \sin 2t)$. At $t = 0$, it was at the origin and its initial velocity is $\mathbf{v}(0) = (1, 0, 0)$. The position function of the particle is ().

5. (2018年期末) If $\mathbf{r}$ is a differentiable vector function of t of constant length, then $\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} = ()$.

6. (2023年期末) The curvature κ for the curve $\mathbf{r}(t) = e^t \cos t \mathbf{i} + e^t \sin t \mathbf{j} + 2\mathbf{k}$ is ().

7. (2021年期中) The maximum curvature κ of function $y(x) = \sin x$ is ().

8. (2023年期中) Let C be the curve given by $\mathbf{r}(t) = \sqrt{2} \cos t \mathbf{i} + \sin t \mathbf{j} + \sin t \mathbf{k}$.

(a) Find the unit tangent vector $\mathbf{T}(t)$ and principal unit normal vector $\mathbf{N}(t)$ for C.

(b) Find the curvature κ .

9. (2018年期末) Find the unit tangent vector $\mathbf{T}$, the principal unit normal vector $\mathbf{N}$, and the curvature κ for the plane curve $\mathbf{r}(t) = (2t + 3)\mathbf{i} + (5 - t^2)\mathbf{j}$.

10. (2021年期中) Assume $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where

$$f(t) = \int_0^t \cos(x^2) dx, \quad g(t) = -t \cos t, \quad h(t) = \sum_{n=1}^{\infty} \frac{t^n}{n}.$$

Calculate $\mathbf{r}'(0)$.

11. (2020年期末) Find the points on the curve $\mathbf{r}(t) = (12 \sin t)\mathbf{i} - (12 \cos t)\mathbf{j} + 5t\mathbf{k}$, at a distance 26π units along the curve from the point $(0, -12, 0)$.

12. (2019年期末) Find the equation of the osculating circle for the parabola $y = x^2$ at $x = 1$.

13. (2019年期中) A particle is located at the point $(1, 0, -2)$. Its initial speed is $|\mathbf{v}(0)| = 3$ at time $t = 0$, and the direction of its initial velocity is toward the point $(2, -1, 3)$. The particle moves with constant acceleration $\mathbf{i} + 2\mathbf{j} + \mathbf{k}$. Find its position vector $\mathbf{r}(t)$ at time t .